R15/R13

Code No: 123AN/113AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2021
PROBABILITY AND STATISTICS
(R15-Common to ME, CSE, IT, MCT, AME, MIE; R13-Common to ME, CSE, IT, MCT, AME, MIE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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Used Papers 08-03-21AM

1.a) The frequency function of a continuous random variable X is given by $f(x) = Ax(2-x), 0 \le x \le 2$.
Find variance of X.

b) Let $f(x) = 3x^2$, when $0 \le x \le 1$ be the probability density function of a continuous random variable X. Determine a and b such that
i) $P(X \le a) = P(X > a)$ ii) $P(X > b) = 0.05$. [7+8]

2.a) Find the moment generating function of $f(x) = \frac{1}{8}(3c_x), x = 0, 1, 2, 3.$ and hence find its mean and variance.

b) If 3 fair coins are tossed and if X is the total number of heads in the outcome. Show that X is a random variable. [7+8]

3. The joint probability density function of the random variable X and Y is given by $f(x, y) = K(xy + y^2), 0 \le x \le 1, 0 \le y \le 2.$
Find $P(Y > 1), P\left(X > \frac{1}{2}, Y < 1\right)$ and $P(X + Y \le 1)$ [15]

4.a) Construct sampling distribution of means for the population 3, 7, 11, 15 by drawing samples of size two with replacement. Determine i) μ ii) σ iii) Sampling distribution of means iv) $\mu_{\bar{x}}$ v) $\sigma_{\bar{x}}$.

b) The mean of certain normal population is equal to the standard error of the mean of the samples of 64 from that distribution. Find the probability that the mean of the sample size 36 will be negative. [8+7]

5.a) What is meant by a statistical hypothesis? What are the two types of errors of decision that arise in testing a hypothesis?

b) A random sample of boots worn by 40 combat soldiers in a desert region showed an average life of 1.08 years with a standard deviation of 0.05 years. Under standard conditions the boots are known to have an average life of 1.28 years. Is there reason to assert at a level of significance of 0.05 that use in the desert causes the mean life of such boots to decrease? [7+8]

6.a) Discuss various types of alternative hypothesis with suitable example.

b) In 1950 in India the mean life expectancy was 50 years. If the life expectancies from a random sample of 11 persons are 58.2, 56.6, 54.2, 50.4, 44.2, 61.9, 57.5, 53.4, 49.7, 55.4, 57.0. Does it confirm the expected view? [7+8]

7.a) Discuss basic queuing process.

b) Workers come to a tool store room to enquiry about the special tools for a particular job. The average time between the arrivals is 60 seconds and the arrivals are assumed to be in Poisson distribution. The average service time is 40 seconds. Find i) average queue length ii) Average length of non-empty queue. [7+8]

8.a) A fair die is tossed repeatedly. If X_n denotes the maximum of the number occurring in the first n tosses, find the transition probability matrix P of the Markov chair $\{X_n\}$. Find also P^2 and $P(X_2 = 6)$.

b) The transition probability matrix of a Markov chain is given by $\begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.2 & 0.8 \end{bmatrix}$. Is this matrix irreducible? [8+7]

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